

Jet Engines

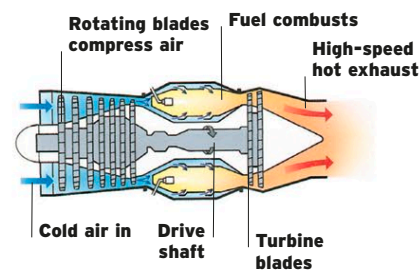
The jet engine had a long gestation and a difficult birth. We now take for granted the principle of compressing air, adding fuel in a combustion chamber and burning it to create thrust that drives an engine forward. In truth, pioneers like the great Frank Whittle faced huge problems in the 1920s and 30s in developing compressors that would deliver an acceptable pressure ratio, combustion chambers that would sustain a flame while air flowed through at huge speed, and turbines

that would not simply melt in the high temperatures created. Among other things, Whittle's turbojet engine required a compressor that delivered twice the pressure ratio of contemporary superchargers. Few people at the time thought the idea would work and government support was denied until it became apparent in the late 1930s that the piston engine was reaching its limits and new high-temperature alloys might just make the turbine engine possible.

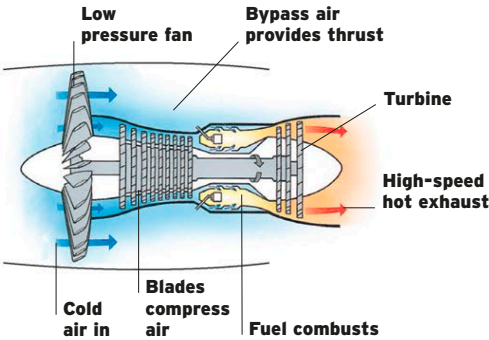
TYPES OF JET ENGINE

The simplest form of gas turbine aero engine is the turbojet, in which all the thrust is provided by the jet leaving the engine at high speed. However, the high-velocity, small-diameter jet of exhaust gases these engines produce is not a particularly efficient means of propulsion, especially at low airspeed. For this reason, turbojets were first used in military aircraft and high-speed airliners. Most modern civil aircraft use turboprop and turbofan engines that use an increased diameter of jet flow. The turboprop uses turbine power to drive a propeller, while the turbofan uses what is, in effect, an oversized compressor stage to provide extra air flow that bypasses the rest of the engine.

Turbojet The turbojet draws in air through a compressor driven by a turbine running in its exhaust flow. Compressed air flows around and into a combustion chamber into which fuel is sprayed. (Once the engine is started, using spark igniters, the flame burns continuously.) The hot exhaust leaves the rear of the engine through a nozzle designed to give maximum velocity.



Turbofan Used for most commercial airliners, a turbofan drives a large ducted fan at the front of the engine. In airliners, the "bypass" flow produced by this fan can be up to ten times the flow passing through the "core" engine. Emerging at low velocity, the combined bypass air/exhaust gives very efficient propulsion and low noise at typical 600-plus mph (965-plus km/h) airliner cruising speed.



Turboprop A turboprop engine drives a propeller by taking power from an additional turbine running in exhaust gas. The exhaust gives some extra thrust but the main thrust is provided by the propeller. In other ways turboprops resemble turbojet engines. They are most efficient at speeds of up to 450 mph (724 km/h).

